Project Code | *Kod Projek*	**BAXU 3973**

UNIVERSITI TEKNIKAL MALAYSIA MELAKA
FACULTY OF ARTIFICIAL INTELLIGENCE AND CYBER SECURITY

PROJEK SARJANA MUDA 1: PROPOSAL FORM

A TITLE OF PROPOSED PROJECT | *TAJUK PROJEK YANG DICADANGKAN*

AI-POWERED REAL-TIME POTHOLE DETECTION AND HAZARD RISK ASSESSMENT SYSTEM FOR MICRO-MOBILITY (BICYCLE AND ELECTRIC SCOOTER) USERS

B DETAILS OF STUDENT | *BUTIRAN PELAJAR*

Name	MUHAMMAD AL IEQBAL NUR HAQIEMI BIN ABDUL KHALID			**Program:**	**BAXI** [/] **BAXZ** [ ]
Matric No.	B032310536				
Handphone No.	012-795 6167				
Semester	2	**Session**	2025/2026	**Email Address**	B032310536@student.utem.edu.my

C PROJECT INFORMATION | *MAKLUMAT PROJEK*

(i) Executive Summary of Project Proposal [Maximum 300 words]

Micro-mobility such as bicycles and electric scooters has become a cornerstone of modern urban transport and short-distance commuting. However, these vehicles are uniquely vulnerable to road hazards. Unlike other transportation such as motorcycles or cars, the small wheel diameter and often rigid suspension of both bicycles and e-scooters mean that even minor potholes can cause immediate loss of control or mechanical failure. While GPS apps provide route guidance, they lack real-time data regarding road surface quality. This project proposes a solution to bridge safety.

The system utilizes a smartphone camera mounted on the handlebars to detect potholes using a highly optimized YOLOv8 Nano model. To ensure zero-latency detection and preserve user privacy, the artificial intelligence runs completely offline via on-device Edge Computing (TensorFlow Lite). A key innovation of the system is the integration of a Risk Assessment Algorithm that factors in the vehicle's real-time velocity (via GPS) and the detected hole's size to calculate an "Accident Risk Percentage." Because micro-mobility vehicles are highly sensitive to surface changes, the system provides high-frequency audio-visual alerts to allow users sufficient time to brake or swerve.

(ii) **Detailed Proposal of the Project**

(a) Introduction *(Project Background and Problem Statements)*

Background: Micro-mobility is an eco-friendly transport trend in Malaysia, yet road maintenance often overlooks the specific needs of these smaller vehicles. Bicycles and e-scooters are highly susceptible to "surface-induced accidents" where even a small hole can throw a rider off-balance.

Problem Statement: Existing safety technology is largely designed for larger motor vehicles. For e-scooter and bicycle riders, there is no real-time proactive warning system that identifies hazards.

(b) Objectives of the Project

This project embarks on the following objectives:

1. To develop and train a lightweight Object Detection model (YOLO) capable of identifying potholes in real-time on a mobile device.
2. To implement a Risk Assessment Algorithm that calculates accident probability based on pothole dimensions and real-time vehicle velocity.
3. To design a User Interface for a mobile application that delivers immediate, non-intrusive safety warnings via audio and visual cues.

(c) Scope of the Project

- **Target User:** Urban commuters using bicycles and electric scooters.
- **Platform:** Mobile application (Android) using TensorFlow Lite.
- **Environment:** Dedicated bike lanes, urban roads, and park pathways during daylight.
- **Technology:** YOLOv8/v11 Nano, Flutter/Kotlin, and GPS API for speed tracking.

(d) Expected Outcome/ Proposed Solution

- A trained TFLite model with high precision and recall for road hazards.
- An Android application that demonstrates real-time detection and risk-level warnings.
- A technical report detailing the correlation between speed, hole size, and rider safety.

D	**REFERENCES** | *RUJUKAN*
State your references (Minimum 10 references)	
1	*Redmon, J., & Farhadi, A. (2018). YOLOv3: An Incremental Improvement.*
2	*Arya, D., et al. (2020). Deep Learning-based Road Damage Detection and Classification for Multiple Countries.*
3	*Ukhwah, N. N., et al. (2021). Pothole Detection Implementation using YOLOv4.*
4	*Ma, N., et al. (2018). ShuffleNet V2: Practical Guidelines for Efficient CNN Architecture Design.*
5	*Basavaraju, A., et al. (2022). Real-time Pothole Detection using Deep Learning.*
6	*Official Malaysia Road Safety Department (JKJR) Statistics on Motorcycle Accidents.*
7	*Kim, Y. M., et al. (2021). Deep Learning-Based Pothole Detection Using Smartphone Sensors.*
8	*TFLite Documentation for Mobile Vision Tasks.*
9	*Analysis of Kinetic Energy Impact in Motorcycle-Road Hazard Collisions.*
10	*Comparative Study of YOLOv8 vs. YOLOv11 for Edge Computing.*

E	**DECLARATION BY STUDENT** | *AKUAN PELAJAR*		
(i)	**Date:** 13/2/2026	**Student's Signature:**	

E	**RECOMMENDED BY SUPERVISOR** *PERAKUAN OLEH PENYELIA*	
	Recommended	☐
	Not Recommended	☐
	Comments:	
	Supervisor's Name:	
	WAN MOHD YA'AKOB BIN WAN BEJURI, DR	

	Signature & Stamp:
	Date: